2c. Content of chapters C1 to C8

Chapter C1: Air and water

Overview

The quality of our air and water is a major world concern. Chemists monitor our air and water, and work to minimise the impact of human activities on their quality.

In Topic C1.1, the context of changes in the Earth's atmosphere is used to explore the particle model and its limitations when explaining changes of state, and the principles of balancing equations for combustion reactions. As a development of ideas about burning fuels, Topic C1.2 considers bonding in small molecules and temperature changes in chemical reactions.

Topic C1.3 explores the evidence for climate change, asking why it might be occurring and how serious a threat it is. Learners consider environmental and health consequences of some air pollutants and climate change, and learn how scientists are helping to provide options for improving air quality and combatting global warming.

Finally, Topic C1.4 explores the need for increasing the amount of potable water worldwide, and techniques for obtaining potable water from ground, waste and salt water.

Learning about air and water before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- be able to explain the properties of the different states of matter (solid, liquid and gas) in terms of the particle model, including gas pressure
- appreciate the differences between atoms, elements and compounds
- be familiar with the use of chemical symbols and formulae for elements and compounds
- know about conservation of mass, changes of state and chemical reactions
- be able to explain changes of state in terms of the particle model
- know that there are energy changes on changes of state (qualitative)
- know about exothermic and endothermic chemical reactions (qualitative)
- understand the carbon cycle
- know the composition of the Earth's atmosphere today
- know about the production of carbon dioxide by human activity and its impact on climate.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about air and water at GCSE (9–1)

C1.1 How has the Earth's atmosphere changed over time, and why?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
The Earth, its atmosphere and its oceans are made up from elements and compounds in different states. The particle model can be used to describe the states of these substances and what happens to the particles when they change state. The particle model can be represented in different ways, **but these are limited because they do not accurately represent the scale or behaviour of actual particles, they assume that particles are inelastic spheres, and they do not fully take into account the different interactions between particles**. The formation of our early atmosphere and oceans, and the state changes involved in the water cycle, can be described using the particle model.	1. recall and explain the main features of the particle model in terms of the states of matter and change of state, distinguishing between physical and chemical changes and recognise that the particles themselves do not have the same properties as the bulk substances 2. **explain the limitations of the particle model in relation to changes of state when particles are represented by inelastic spheres** 3. use ideas about energy transfers and the relative strength of forces between particles to explain the different temperatures at which changes of state occur
Explanations about how the atmosphere was formed and has changed over time are based on evidence, including the types and chemical composition of ancient rocks, and fossil evidence of early life (IaS3). Explanations include ideas about early volcanic activity followed by cooling of the Earth resulting in formation of the oceans. The evolution of photosynthesising organisms, formation of sedimentary rocks, oil and gas, and the evolution of animals led to changes in the amounts of carbon dioxide and oxygen in the atmosphere.	4. use data to predict states of substances under given conditions 5. interpret evidence for how it is thought the atmosphere was originally formed 6. describe how it is thought an oxygen-rich atmosphere developed over time

Linked learning opportunities

Practical work:
- measure temperature against time and plot a cooling curve for stearic acid or heating curve for ice

Ideas about Science:
- use the particle model to explain state changes (IaS3)
- distinguish data from explanatory ideas in accounts of how the atmosphere was formed (IaS3)

C1.1 How has the Earth's atmosphere changed over time, and why?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Our modern lifestyle has created a high demand for energy. Combustion of fossil fuels for transport and energy generation leads to emissions of pollutants. Carbon monoxide, sulfur dioxide, nitrogen oxides and particulates directly harm human health. Some pollutants cause indirect problems to humans and the environment by the formation of acid rain and smog. Scientists monitor the concentration of these pollutants in the atmosphere and strive to develop approaches to maintaining air quality (IaS4).	7. describe the major sources of carbon monoxide and particulates (incomplete combustion), sulfur dioxide (combustion of sulfur impurities in fuels), oxides of nitrogen (oxidation of nitrogen at high temperatures and further oxidation in the air)	**Ideas about Science:** • unintended impacts of burning fossil fuels on air quality (IaS4) • catalytic converters, low sulfur petrol and gas scrubbers as positive applications of science (IaS4)
	8. explain the problems caused by increased amounts of these substances and describe approaches to decreasing the emissions of these substances into the atmosphere including the use of catalytic converters, low sulfur petrol and gas scrubbers to decrease emissions	
The combustion reactions of fuels and the formation of pollutants can be represented using word and symbol equations. The formulae involved in these reactions can be represented by models, diagrams or written formulae. The equations should be balanced. When a substance chemically combines with oxygen it is an example of oxidation. Combustion reactions are therefore oxidation. Some gases involved in combustion reactions can be identified by their chemical reactions.	9. use chemical symbols to write the formulae of elements and simple covalent compounds	
	10. use the names and symbols of common elements and compounds and the principle of conservation of mass to write formulae and balanced chemical equations	
	11. use arithmetic computations and ratios when balancing equations M1a, M1c	
	12. describe tests to identify oxygen, hydrogen and carbon dioxide *PAG2*	
	13. explain oxidation in terms of gain of oxygen	